

INVITED ARTICLE

Isotretinoin updates

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ABSTRACT: Acne vulgaris is a chronic inflammatory disease of pilosebaceous follicles commonly affecting adolescents and young adults. This disease has a profound psychological impact on affected individuals and treatment has been shown to significantly improve both self-esteem and quality of life. Isotretinoin is an effective medication used primarily in severe cystic acne patients. Over the past 30 years, this medication has revolutionized the treatment of acne. However, despite its popularity there are numerous side effects associated with its use. Most of its side effects are predictable and dose dependent, which has led to the development of variable dose regimens. Unfortunately, rare but significant side effects (e.g., depression, inflammatory bowel disease) do occur and necessitate careful monitoring to improve clinical outcomes and minimize potential adverse events.

KEYWORDS: acne, isotretinoin, updates

Introduction

Acne vulgaris is a chronic inflammatory disease of pilosebaceous follicles commonly affecting adolescents and young adults. An estimated 75–95% of all teenagers suffer from acne to some degree (1,2). Skin with the highest density of hormonally responsive sebaceous follicles is most commonly affected. This includes the face, neck, upper trunk, back and upper arms. Clinically, patients present with open and closed comedones, inflammatory papules, pustules, cysts and nodules, and in some cases permanent scarring (3). Both genetic and environmental factors play a role in acne pathogenesis (4,5). The disease has a significant impact on quality of life, and treatment has been shown to significantly improve self-esteem (6).

The pathogenesis of acne is multifactorial, caused by a combination of follicular hyperkeratinization, sebum production, *Propionibacterium acnes* (*P. acnes*) colonization, and

inflammation. It has been traditionally believed that follicular hyperkeratinization with microcomedone development was the initiating factor in acne. However, recent evidence illustrates that inflammatory events occur prior to microcomedone formation, challenging this notion. Jeremy et al. demonstrated a perifollicular increase in CD3+ and CD4+ T cells, macrophages, and interleukin (IL)-1 in clinically normal appearing skin of acne patients, but not in nonacne control patients (7). Subclinical inflammation may be the inciting factor leading to the microcomedone development to begin with.

The sebaceous gland plays a pivotal role in acne pathogenesis. Androgen stimulation of sebaceous glands leads to an increase in sebum, which provides a favorable environment for *P. acnes*' growth. Lipases produced by *P. acnes* break down sebum into free fatty acids, which cause follicular inflammation. Additionally, *P. acnes*' heat shock proteins, porphyrins, and other enzymes (e.g., lipase, protease, hyaluronidase) (8,9) cause tissue damage. *P. acnes* stimulates the innate immune system through activation of toll-like receptor-2 (TLR-2) on perifollicular macrophages, leading to monocyte secretion of proinflammatory cytokines (IL-8, IL-12). This inflammation may result in

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follicular rupture. *P. acnes* has also been demonstrated to increase in matrix metalloproteinase (MMP) mRNA expression, and these MMP's lead to dermal matrix degradation and play a key role in scar formation (10).

The armamentarium of therapies for acne vulgaris currently available includes systemic and topical retinoids, systemic and topical antimicrobials, hormonal therapies for females, topical dapsone, alpha and beta hydroxy acids, azelaic acid, as well as laser and light-based therapies. These therapies are commonly used in combination to address simultaneously as many pathogenic factors as possible. Only oral isotretinoin simultaneously addresses all of the four major pathogenic factors at the same time.

Basic science

Isotretinoin, *aka* 13-*cis*-retinoic acid, is a vitamin A derivative naturally present in very small quantities in the blood and tissues of humans. It is relatively water-soluble and metabolized in the liver by the cytochrome p450 system. In the liver, isotretinoin is oxidized to 4-*oxo*-isotretinoin and isomerized into several metabolites including tretinoin. Maximum plasma concentrations of isotretinoin are reached between 1 and 4 hours following oral administration (11). By 6–20 hours, peak levels of 4-*oxo*-isotretinoin are seen in the plasma. Retinoids levels return to baseline after 1 month of discontinuing oral isotretinoin. The drug's metabolites are excreted primarily in the feces (53–74%) and small amounts in the urine (11).

Retinoids impact all of the major pathogenic factors in acne (12,13). Isotretinoin acts as a pro-drug (14,15). Intracellularly, it is converted into five major active metabolites ((1)13-*cis*-4-*oxo*-retinoic acid/4-*oxo*-isotretinoin, (2) all-*trans*-retinoic acid/tretinoin, (3) all-*trans*-4-*oxo*-retinoic acid/4-*oxo*-tretinoin, (4) 9-*cis*-retinoic acid, and (5) 9-*cis*-4-*oxo*-retinoic acid). These bind nuclear retinoic acid receptors (RARs or RXRs) which mediate retinoid activity. RARs and RXRs form heterodimers to regulate gene networks responsible for cell proliferation, differentiation, survival, and death (16). There are three retinoic acid A receptors (RAR alpha, beta, and gamma) and three retinoic acid X receptors (RXR alpha, beta, and gamma), but the RAR gamma and RXR alpha subtypes are most commonly expressed in the skin (17). Isotretinoin isomerizes into tretinoin and then binds these retinoid receptors to exert its anti-proliferative effect on sebocytes, consequently inhibiting cell

differentiation, sebum secretion, and sebaceous gland size. Sebocyte apoptosis may occur independent of RAR activation (18).

Isotretinoin use results in reduction of *P. acnes* colonization and inflammation. A decrease in sebum production with shrinkage of sebaceous glands duct creates an unfavorable environment for *P. acnes* growth (19–21). This results in a reduction in proinflammatory mediators from *P. acnes* as well as inhibition of polymorphonuclear leukocyte chemotaxis (22,23). Moreover, isotretinoin has been demonstrated to downregulate surface expression of TLR-2 on circulating monocytes, explaining the blunted inflammatory cytokine response to *P. acnes* (24).

Indication

During the 1960–1980s, the use of retinoids was studied in various skin diseases, ranging from psoriasis to skin cancer. Its efficacy in acne was discovered in the 1970s (25,26). It was not until 1982 that the Food and Drug Administration (FDA) approved oral isotretinoin for the treatment of severe, recalcitrant nodular acne unresponsive to conventional therapy, including systemic antibiotics. Despite having only an acne indication, isotretinoin has been reported to be used off-label for various dermatoses. These include gram-negative folliculitis, inflammatory rosacea, and pyoderma faciale/rosacea fulminans (27). Its efficacy has also been described in disorders of keratinization such as pityriasis rubra pilaris, Darier's disease, ichthyoses, and keratodermas (28). Isotretinoin has also been evaluated as chemoprevention for cutaneous neoplasms (29) and as adjuvant therapy in various genodermatoses like xeroderma pigmentosum and epidermodysplasia verruciformis (30–32). Isotretinoin suppresses growth of malignancies only while actively used, the malignancy may return after the drug is discontinued.

Clinical efficacy

In most patients, isotretinoin therapy yields skin improvement by the third month of treatment (33). About 60–85% of patients are expected to achieve a clinical cure rate after 20 weeks of treatment (34–36). Layton et al. observed that 61% of patients were acne-free after one course of treatment, 16% required a second isotretinoin course, and 23% required oral antibiotics after their isotretinoin treatment (37). In a retrospective study, 76.8% of

patients were acne-free 2 years after a single course of isotretinoin, and the remaining 23.2% required further medical management (38).

The majority of relapses occur within 3 years of completing isotretinoin therapy (39). Up to 65% of relapses may be attributed to subtherapeutic cumulative isotretinoin doses, defined as less than 120 mg/kg in total dose (40–51). In addition, intermittent dosing regimens, male gender, patients with predominantly truncal acne, and patients less than 16 years of age are also risk factors for relapse (52,53). One study directly comparing patients receiving a higher versus lower dose demonstrated a relapse rate of 82% in patients who received less than 120 mg/kg, compared to 20% in those who received greater than 120 mg/kg (54). In addition to cumulative levels of the drug, dosing regimen may influence relapse rates as well. Lee et al. found a significantly higher relapse rate in patients on an intermittent dosing regimen (taking drug 1 week out of every four), compared to those continuously taking the drug, at both high and low doses (55). While more data is needed to understand relapses more clearly, what is known is that when relapses do, the acne is generally more responsive to conventional treatments than the initial outbreak (53).

Contraindications and drug interactions

A careful concomitant drug review and discussion of the drug's risks should be performed prior to initiating isotretinoin therapy. Tetracycline class antibiotics should not be administered with isotretinoin because of an amplified risk of pseudotumor cerebri from both drugs. Vitamin A supplements should be avoided due to additive toxic effects of isotretinoin. Metabolized by the liver's cytochrome P450 system, caution should be taken when prescribing isotretinoin with other similarly metabolized medications. For example, isotretinoin has been shown to reduce carbamazepine levels (56). Moreover, ketoconazole has been shown to increase isotretinoin plasma levels (53). Isotretinoin is 99.9% bound to albumin in plasma. Medications such as salicylic acid and indomethacin have a high affinity for albumin may displace isotretinoin from albumin and can increase the unbound fraction and activity of the drug (57,58). Patients with known hypersensitivity to isotretinoin or any of its components (e.g., parabens, soy, or other retinoids) should not be treated with isotretinoin (53). Finally, heavy alcohol

intake should be avoided, as it has been shown to reduce efficacy of the drug (59).

All women of childbearing potential are required to use two effective forms of contraception simultaneously because major fetal abnormalities have been documented related to isotretinoin administration. Pregnancy, planning pregnancy, and breast-feeding are absolute contraindications to therapy (53). Moreover, patients should not donate blood during the course of therapy or 1 month after discontinuation, because of the risk of exposing pregnant women or women planning pregnancy to excessive isotretinoin levels during a blood transfusion (60). More details regarding pregnancy will be covered in the "Teratogenicity and the iPLEDGE Program" section of this paper.

The patient's medical history must be evaluated to determine whether he/she is an appropriate isotretinoin candidate. Caution should be exercised if considering isotretinoin in patients with psychiatric disorders and suicidal ideation, as there are conflicting reports on exacerbation of psychiatric disorders associated with isotretinoin. Other preexisting medical conditions that should be considered include diabetes mellitus, anorexia nervosa, hyperlipidemia, inflammatory bowel disease, skeletal disorders, seizure disorders, and pancreatitis. More details will be covered in the "Side Effects" section of this paper.

Dosing

Isotretinoin is a lipophilic molecule which is optimally absorbed when taken with a fatty meal. When taken without food, fasted plasma levels can be nearly 60% lower than fed levels (61). In late 2012, a new branded isotretinoin formulation was released into the US market. This drug is formulated in such a way that the isotretinoin molecule is incorporated into a lipid-based matrix, enhancing the absorption of isotretinoin independent of meals (62). The half life of isotretinoin is 10–20 hours, and a twice daily dosing regimen is therefore ideal. In fact, studies have shown that when dosed twice daily drug trough *exceed* drug peak levels when dosed once daily (58).

The generally recommended dosage of isotretinoin ranges between 0.5 and 2.0 mg/kg/day to reach a target cumulative dose of 120–150 mg/kg. Therapy is typically initiated at 0.5 mg/kg/day for the first month, then increased as tolerated to 1 mg/kg/day for the duration of therapy (63). When prescribed this way, a treatment course lasts

approximately 20 weeks; however, longer courses may be used for more severe cases (64). Lower daily doses may be chosen for older patients, those who cannot tolerate side effects, or for patients in whom higher doses may result in an acne flare (65). In these cases, lower daily doses may be given over longer periods of time to achieve the same cumulative dosage. Patients who do not reach the cumulative target dose (e.g., who discontinue therapy prematurely) are more likely to relapse (66). While the majority of patients completing a full, single course achieve a sustained remission, some patients do require additional treatment. In this case, a second course is typically initiated no sooner than 8 weeks after completing the first course (67).

Low-dose regimens may be preferred because of a more favorable side-effect profile and improved patient adherence (68–70). One review of several studies conducted between 1987 and 2009 found no statistical differences in the efficacy (88.52% vs. 87.8%; $p > 0.05$) and relapse rates (21.48% vs. 34.6%; $p > 0.05$) of patients treated with low-dose isotretinoin versus high dose (71–77). In addition, Amichai et al. found that patients on 0.3–0.4 mg/kg/day for 6 months had complete remission in 93–95% of patients and reduced relapse rate of 4–6% (69).

Similar to low-dose regimens, intermittent or variable dosing may offer patients a favorable side-effect profile and be more cost effective in some cases. Goulden et al. published a study in which adult patients with acne unresponsive to traditional treatments were placed on an intermittent regimen of isotretinoin dosed at 0.5 mg/kg/day for 1 week out of every 4 weeks, for 6 months. At the end of treatment, 88% of patients achieved complete clearance. During the 12-month follow-up period, 39% of patients relapsed (73). Kaymak et al. evaluated patients with mild to moderate acne treated with isotretinoin dosed at 0.5–0.75 mg/kg/day for 1 week out every 4 weeks, for a total period of 6 months. At the end of the study, 82.9% of patients achieved complete clearance (72,78).

Isolated studies have evaluated the efficacy and safety of high-dose isotretinoin regimens. Many isotretinoin side effects are dose dependent, making higher doses challenging to use. One retrospective chart review revealed that higher doses of isotretinoin therapy resulted in a reduced frequency of relapse and retreatment, as compared to lower daily doses (51). Following patients for 10 years, Cunliffe et al. found that patients dosed at 1 mg/kg/day had a lower recurrence rate (22–30%)

than patients dosed at 0.5 mg/kg/day (39–82%) (78). Another study found that higher daily (average of 1.6 mg/kg/day) and cumulative (average of 290 mg/kg) doses were safe and effective. Over the 3-year study period, 10 patients (12.5%) required a second course of isotretinoin (79).

Limited data exist that directly compare different isotretinoin dosing regimens. One study by Lee et al. in South Korea found that high and low dose continuous regimens were comparable, and both resulted in superior efficacy compared to intermittent dosing (61). After 1 year, both continuous therapy groups had equivalent relapse rates. Patients in the lower dose group reported higher levels of satisfaction with the regimen. Patients in the intermittent therapy group were least satisfied with their treatment, had a higher rate of relapse, and lesser degree of acne improvement (61). Another study compared two different intermittent dosing regimens to a continuous therapy regimen. In one arm, patients were treated with isotretinoin at 0.5 mg/kg/day for the first 10 days of each month, in another arm patients received isotretinoin 0.5 mg/kg/day for the first month followed by 0.5 mg/kg/day first 10 days of each additional month. In the third arm, patients received continuously dosed isotretinoin at 0.5 mg/kg/day for 6 months. The investigators concluded that both intermittent regimens were as effective as the continuous dosage. While the number and severity of side effects were lower in both of intermittent groups compared to the continuous group, the relapse rate at 12-month follow-up was higher (80).

When initiating isotretinoin therapy in a patient with severe nodulocystic acne, caution must be exercised to avoid a flare which could result in permanent scarring. In some cases, acne actually worsens with initial isotretinoin therapy. For this reason, it is recommended that patients are started on a lower dose of isotretinoin then escalated as tolerated. Oral corticosteroids (e.g., prednisone dosed at 0.5–1.0 mg/kg/day for 2–6 weeks) may be administered at the same time as initiating isotretinoin to prevent this flare. The corticosteroid should be slowly tapered down within a 2–4-week period. Alternatively, patients may be pretreated with 1 to 2 weeks of systemic corticosteroids prior to starting isotretinoin. Predictive factors for acne flare during isotretinoin treatment include male gender, and the presence of macrocomedones, truncal comedones, severe acne with facial nodules, and a high number of facial comedones (81–83).

Teratogenicity and the iPLEDGE program

While safe when prescribed and monitored appropriately, isotretinoin is associated with serious side effects. Isotretinoin is a potent teratogen. A developing fetus is most susceptible to the effects of isotretinoin during organogenesis in the first trimester. It is estimated that 18–22% of pregnancies exposed to isotretinoin result in spontaneous abortion (84,85). Of those fetuses that survive, 20–30% will develop malformation including craniofacial defects (cleft lip, anotia, small/absent auditory canals, microphthalmia), cardiac abnormalities (septal defects), central nervous system abnormalities (microcephaly, hydrocephalus), thymic hypoplasia, and limb reduction or duplication (86,87). While data is limited, fetuses exposed to isotretinoin but born without anatomical birth defects are thought to have lower levels of neuropsychological functioning and developmental delays later in life (87). Isotretinoin is FDA pregnancy category X, demonstrating a clear risk of teratogenicity, where risks outweigh benefits of use.

Isotretinoin was initially introduced to the US market place in 1982, after which time prescription rates rose rapidly. Since then, several risk management programs were developed to control fetal exposure to isotretinoin. This has ultimately resulted in the current iPLEDGE Program. The drug company Hoffman-La Roche, who marketed the initial brand name Accutane®, first developed a voluntary outreach program providing educational brochures about the drug's teratogenic effects. Hoffman-La Roche later launched the Accutane Pregnancy Prevention Program (PPP), which was in place from 1988 to 2001. Patients were required to sign a consent form, have a negative pregnancy test before initiating isotretinoin, and be counseled on using two forms of birth control (88). In conjunction with the FDA, Hoffman-La Roche's PPP program was abandoned for 2002 SMART program ("System to Manage Accutane-Related Teratogenicity"). The SMART program required registration of all isotretinoin users, restricted prescribing privileges to providers who successfully completed educational and consent information programs, required contraception counseling, and pregnancy monitoring (89).

Around the same time that the SMART program was implemented, several generic isotretinoin drugs were released into the US market. Each of the pharmaceutical companies launched separate monitoring programs, including SPIRIT (the

System to Prevent Isotretinoin-Related Issues of Teratogenicity), IMPART (the Isotretinoin Medication Program Alerting you to the Risks of Teratogenicity), and ALERT (the Adverse Event Learning and Education Regarding Teratogenicity), respectively (90). Multiple risk management programs created conflict, and in 2006, the FDA approved a single system, known as iPLEDGE, which is the program currently in place. iPLEDGE is a comprehensive, computer-based program that strictly monitors the prescribing, dispensing, and distribution of isotretinoin. All patients (females of childbearing potential, females who are not of childbearing potential, and men) are required to be registered and visit their provider monthly for monitoring. While there are no reported birth defects in babies fathered by men on isotretinoin, it is still recommended that men should not father during therapy (38,91). Women of childbearing potential have more requirements to fulfill in iPLEDGE than men or women not of childbearing potential. These include: (i) the use of two forms of contraception while on isotretinoin therapy for at least 1 month prior to starting isotretinoin, during therapy and for 1 month post-therapy; (ii) completion of a monthly assessment evaluating their understanding of the fetal risks involved in isotretinoin exposure; (iii) two negative pregnancy tests (urine or blood) must be conducted prior to receiving the initial prescription for isotretinoin as well as one negative pregnancy test monthly prior to receiving their refills; (iv) patients must fill their prescription within 7 days of the negative pregnancy test (90). Additionally, any woman who becomes pregnant while on isotretinoin is required to enroll in the iPLEDGE pregnancy registry.

Adverse events

Mucocutaneous adverse events are common and usually dose dependent. Chelitis is common, developing in up to 98% of patients (13). In fact, the lack of chelitis may be a marker for noncompliance with therapy (90). Rademaker et al. found the higher the dose of isotretinoin, the higher the prevalence of chelitis (92). Xerosis, with occasional accompanied pruritus, occurs in up to 50% of patients (13), and may be especially prevalent in patients with an atopic diathesis (39). Mucocutaneous reactions can be controlled with regular use of artificial tears, moisturizers/bland emollients and lip balms. Other mucocutaneous side effects include but are not limited to: acral desquamation, facial erythema, eczema (which

may require topical corticosteroid therapy), skin fragility, paronychia, pyogenic granulomas, bruising, eruptive xanthomas, photosensitivity (due to thinning of the stratum corneum), paronychia, onycholysis, nail plate fragility, and alopecia (13,38,53,93). Other side effects are relatively uncommon but vast and are listed in Table 1.

Lipid abnormalities

Lipid abnormalities are not an uncommon side effect during treatment with isotretinoin. It is estimated that 20–45% of patients develop hypertriglyceridemia, while 30% develop elevated total cholesterol and low-density lipoprotein levels. In some cases, there may be a decrease in high-density lipoprotein levels (93). Risk factors for isotretinoin-induced lipid abnormalities have been identified. These include truncal obesity (increased waist-to-hip ratio), hyperinsulinemia, presence of apoE gene, and a parent with hypertriglyceridemia (92). Moreover, patients who develop lipid abnormalities during isotretinoin therapy are at risk for metabolic syndrome in the future (88). A personal history hypertriglyceridemia, diabetes, and alcohol consumption likely also play a role in the hyperlipidemia experienced during isotretinoin therapy (94).

Lipid elevations are rarely severe enough to require discontinuation of therapy. They typically occur during the first 1–2 months of treatment and normalize to baseline 2–4 weeks after discontinuation of the drug. Treatment of moderate triglyceride elevation (e.g., 300–500 mg/dL) includes lifestyle changes (weight loss, increased exercise, healthy diet) and/or pharmacologic therapy (e.g., gemfibrozil 600 mg twice daily) (53). In patients who develop triglyceride levels greater than 500 mg/dL, reduction of isotretinoin dose

or cessation of therapy is recommended (53). If triglyceride levels exceed 800 mg/dL, isotretinoin should be immediately discontinued (53). Acute risks of hypertriglyceridemia include the development of pancreatitis and eruptive xanthomas (95). Experts suggest that patients' lipids be monitored frequently, especially those in high-risk categories (96). In the author's practice, patients' lipid panels are monitored monthly with every visit to the office.

Inflammatory bowel disease

The association between isotretinoin use and inflammatory bowel disease (IBD) has long been debated, but a definitive, causal relationship has not been established. Since the 1980s multiple cases have been reported implicating isotretinoin as the cause of patients' IBD flare-ups (97–100). Several large studies evaluating this association have been published to date. In 2009, Bernstein et al. published results of a study using data from the Canadian population epidemiology database. Investigators found no association between isotretinoin and inflammatory bowel disease, either ulcerative colitis or Crohn's disease (101). A second, more recent Canadian study found a small association between isotretinoin and IBD in a teenage population. Moreover, their data suggested a possible association between acne itself with IBD (102). Finally, a 2013 study of 45,000 women of reproductive age on isotretinoin also did not suggest an increase in the risk for IBD, either ulcerative colitis or Crohn's disease (103). On the other hand, in 2010 a large case-control study using a large US insurance claims database found that ulcerative colitis, but not Crohn's disease is associated with previous isotretinoin exposure. The risk of developing ulcerative colitis was associated with

Table 1. Uncommon adverse events associated with isotretinoin (13,37,38,53,93,94)

General abnormalities: headache, nausea, diarrhea, abdominal pain, vomiting, lymphadenopathy, fatigue, weight loss, dizziness, drowsiness, insomnia, lethargy, malaise, nervousness, voice alterations
Liver abnormalities: hepatotoxicity, hepatitis
Skin abnormalities: abnormal wound healing, formation of granulation tissue, pyogenic granulomas, urticaria
Ophthalmic abnormalities: vision impairment, corneal opacities, decreased night vision, keratitis, photophobia, cataracts, contact lens intolerance, blepharoconjunctivitis, xerophthalmia and ocular sicca, optic neuritis, myopia
Renal abnormalities: glomerulonephritis due to drug hypersensitivity
Urogenital abnormalities: proteinuria, hematuria
Musculoskeletal/rheumatologic abnormalities: cutaneous vasculitis, myalgia, arthralgia, rhabdomyolysis, sacroiliitis, elevated creatinine phosphokinase levels, Achilles tendonitis, calcification of ligaments and bones
Otic abnormalities: high tone deafness, tinnitus
Hematologic abnormalities: neutropenia, agranulocytosis, anemia, thrombocytopenia, vascular thrombotic disease
Neurologic abnormalities: pseudotumor cerebri, stroke

higher doses of isotretinoin and longer duration of use of the drug. The absolute risk, however, was very small (104).

Despite a questionable association with ulcerative colitis, there is data in the literature supporting its safe use in this population of patients. Patients with concurrent pyoderma faciale and both Crohn's disease or ulcerative colitis have been successfully and safely treated with isotretinoin, without exacerbation of their gastrointestinal disease (105,106). Moreover, one case series reported the treatment of patients with IBD and severe acne with isotretinoin. Of the four patients, two experienced no adverse events, one patient developed rectal bleeding from hemorrhoids, and one patient had a flare of Crohn's disease (107).

The association between isotretinoin and IBD is further clouded by several confounding factors. Patients present with acne and IBD at a similar age, usually developing in the second decade of life, with a peak in the third decade. Some theorize that the discovery of IBD in patients with a history of isotretinoin use may be incidental rather causal (53). Additionally, severe acne itself may be associated with IBD, independent of isotretinoin (108,109).

Despite a lack of definitive evidence of a causal relationship, the FDA added IBD to the isotretinoin package insert in 2005 as a possible adverse effect. In 2010, the American Academy of Dermatology published a position statement addressing the issue of isotretinoin and IBD: "Current evidence is insufficient to prove either an association or a causal relationship between isotretinoin use and inflammatory bowel disease (IBD) in the general population." (110)

Depression and suicide

Isotretinoin has been associated with the development of mood changes, although a causal relationship has not been established. The medical literature has been flooded with case reports and series of patients with new onset depression or suicidal thoughts during or after treatment with isotretinoin (111–118). In 2003, Hull et al. reviewed all of the psychiatric adverse events reported from the 5 million patients on isotretinoin between January 1966 and May 14, 2003. They found a total of 24 cases of isotretinoin-associated depression, four suicides, and three attempted suicides from isotretinoin overdose (119).

Isotretinoin intake may influence brain activity, explaining patients' altered mood during treat-

ment. In one mouse study, Crandall et al. detected a reduction in hippocampus volume along with diminished neurogenesis, associated with memory and learning impairment associated with isotretinoin intake (120). Similar changes are seen in patients with depression, with more significant changes correlating with depression severity (121,122). Furthermore, several rat and mice models have shown that retinoids cross the blood-brain barrier and bind dopamine receptors. This has been shown to alter the mesolimbic dopamine signaling pathway. This same pathway has been implicated in mood and thought disorders such as addiction, schizophrenia, and depression (123,124). Finally, Bremner et al. demonstrated decreased activity in the orbitofrontal cortex of the brain in patients who completed 4 months of isotretinoin therapy. This is an area of the brain implicated in the pathophysiology of major depression (125).

Despite the body of evidence showing the negative impact of isotretinoin on brain function, there is equal or an even greater amount of evidence refuting the relationship. Studies in rats treated with isotretinoin showed no effect on anhedonia or behaviors associated with depression (126,127). Both a 2001 review of the literature and MedWatch data from 1982 to 1999 (128) and a similar study analyzing FDA reports of depression and suicidal ideation from 1982 to 2000 (129) did not find a clear causal relationship between isotretinoin use and depression or suicide. In a prospective, controlled cohort study in Canada, Cohen et al. evaluated 100 patients treated with isotretinoin compared to 59 treated with topical or other oral acne therapies. They concluded that isotretinoin did not appear to be associated with the development of depression (130). Finally, a recent retrospective review in New Zealand evaluated 1743 isotretinoin patients and found no records of suicidal ideation or attempts. Mood changes leading to discontinuation of the drug developed in only 13 patients, eight of which were successfully retreated with isotretinoin without complications (131).

Existing data suggest that isotretinoin treatment for acne actually improves depression, anxiety, and quality of life (132–134). A retrospective, cohort study by Sundstrom et al. found that the risk of attempted suicide in patients receiving isotretinoin therapy was *lower* during the treatment period compared to both the pre- and post-treatment period (135). Moreover, Yesilova et al. demonstrated that isotretinoin treatment yielded a significant improvement in depression, anxiety, and obsessive thoughts in acne patients (136).

The risk of depression and suicide was added to the package insert of isotretinoin in 1988. In November 2010, the American Academy of Dermatology issued a statement on the issue of isotretinoin and depression, as follows: "A correlation between isotretinoin use and depression/anxiety symptoms has been suggested but an evidence-based causal relationship has not been established" (109).

Bony abnormalities

Since 1944, chronic, high-dose retinoid therapy has reported to be associated with a variety of skeletal abnormalities, including diffuse idiopathic skeletal hyperostosis (DISH) syndrome, calcification of ligaments, osteoporosis, premature fusion of epiphyses, and long bone modeling abnormalities (137–141). Long-term use of isotretinoin (for at least 4 years) in diseases of keratinization (e.g., Darier's disease, ichthyoses, and palmoplantar keratoderma) has been shown to influence bone mineral density (142–144). Radiographic evaluation showed skeletal changes including hyperostosis of the cervical and thoracic spine. Short-term therapy, on the other hand, has not been shown to influence bone mineral density (145–148). The development of DISH has been associated with long-term isotretinoin therapy and is characterized by ossification of ligaments, particularly affecting the spine (149). However, a study by Ling et al. recently showed that no acne patients receiving long-term and/or multiple course of isotretinoin developed clinically significant radiographic bony abnormalities (150). Moreover, there is no data to prove that isotretinoin leads to a DISH-like syndrome when used in the short term.

Monitoring

Given the significant number of potentially severe adverse events, close monitoring of patients on isotretinoin is extremely important. Guidelines for monitoring pregnancy is relatively universal; however, recommendations for monitoring other potential adverse may vary. Females of childbearing potential (FOCP) must adhere to regulations from the iPLEDGE program. They must have a negative pregnancy test at the time of initial enrollment, 1 month later at initiation of therapy, monthly thereafter, and at the conclusion of the course of treatment. In addition, FOCP's must agree to use two forms of contraception (94).

The isotretinoin package insert recommends a fasting lipid panel prior to initiation of therapy

and then regularly once the lipid response to isotretinoin is established, which usually occurs by week 4 (94). The risk of hypertriglyceridemia is particularly important, as it is associated with the development of pancreatitis. While the package insert recommends weekly or biweekly testing, many physicians perform a fasting lipid panel at baseline and then monthly thereafter (97,151,152). European guidelines suggest that a fasting lipid profile be performed at baseline, 1 month into therapy and then every 3 months throughout therapy (13).

Other laboratory tests to be monitored include a comprehensive metabolic panel (CMP) and complete blood count (CBC) with differential. The CMP should include liver function tests (LFTs) to monitor for drug-induced hepatotoxicity. The drug's package insert recommends initial weekly or biweekly LFT testing (94). However, variable monitoring recommendations include baseline and monthly tests thereafter, as is performed in the authors' practice. Suggested guidelines for isotretinoin lab safety monitoring are shown on Table 2. Patients on isotretinoin may be challenged to control blood sugar, so particular attention should be paid to glucose level. Rare hematologic effects of agranulocytosis, neutropenia, anemia, thrombocytosis and thrombocytopenia may occur and should be monitored through CBCs. Isotretinoin use may be associated with the development of arthralgias and muscle aches. However, creatine phosphokinase (CPK) increases are uncommon. There has been one report of rhabdomyolysis associated with strenuous physical activity in a patient on isotretinoin (153). Moreover, in one study, 5 of 89 acne patients treated with isotretinoin were found to have elevated CPK levels during therapy. Of those patients, only one had symptomatic muscle aches. The authors concluded that CPK measurement should be limited to patients who experience severe muscle pain (154).

Conclusion

Isotretinoin is a highly effective, well-tolerated acne treatment that has revolutionized the treatment for severe nodulocystic acne. Detailed informed consent, patient counseling, and careful monitoring are of the utmost important to improve clinical outcomes and minimize potential adverse events. The majority of its common side effects are predictable, dose dependent, and manageable. Despite the controversies that exist regarding associations with

Table 2. Suggested lab tests prior, during, and post treatment with isotretinoin

Test	Screen	Week 0 (screen + 19 days)	Week 4 (treatment period)	Week 8 (treatment period)	Week 12 (treatment period)	Week 16 (treatment period)	Week 20 (treatment period)	Week 24 (follow-up)
B-HCG (urine or serum) at prescriber's office	✓							
B-HCG (urine or serum) at CLIA-certified laboratory		✓	✓	✓	✓	✓	✓	✓
Fasting lipid panel		✓	✓	✓	✓	✓	✓	✓
Liver function tests		✓	✓	✓	✓	✓	✓	✓

B-HCG, beta-human chorionic gonadotropin; CLIA, clinical laboratory improvement amendments.

depression and IBD, isotretinoin is a safe medication when prescribed appropriately.

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